

Project 3 — Build a Specialist

Choose Track A (narrative specialisation) or Track B (arithmetic reasoning), fine-tune an appropriate starting checkpoint, and demonstrate the acquired behaviour through controlled evaluation.

Due: 23 May 2027 · **Weight:** 30%

Page summary

Choose one of two specialisation tracks. Define a specific target behaviour, fine-tune an appropriate starting checkpoint, and demonstrate through controlled evaluation whether the model acquired that behaviour without unacceptable regressions.

Both tracks use the same common assessment principles, parameter limit, compute policy, submission package, and expectation of held-out evaluation. Scores are not compared directly across tracks because their task metrics have different meanings.

Key facts

Item	Value
Weight	30%
Marks available	30

Item	Value
Training stage	Task-specific adaptation
Work mode	Individual; choose Track A or Track B
Model size	32M design target; 33.6M absolute maximum
Compute budget	12 T4-equivalent GPU-hours and 1.5×10^{17} training FLOPs
Main report	20 CVPR pages, excluding references and appendix
Due	Sunday 23 May 2027, 23:59 AET — end of Week 12

What you must demonstrate

Declare one measurable capability, justify the starting checkpoint, establish the unchanged-model baseline, train on a strictly separated dataset, perform at least one controlled comparison or ablation, test held-out generalisation, and measure at least one relevant pre-existing capability for regression.

One carefully investigated function is sufficient. Additional functions, a more complex algorithm, chain-of-thought output, or preference optimisation receive no automatic credit. If constrained decoding or post-processing is used, distinguish model-learned behaviour from externally enforced behaviour.

Track A — Narrative Specialist

Continue from your frozen Project 1 model, your frozen Project 2 model, or the course narrative fallback checkpoint, and add at least one specific, testable capability. Using the course fallback carries no mark penalty. Supported examples include integrating specified vocabulary naturally into a continuation, completing a short story under one or two explicit conditions, producing a

specified broad ending type, or detecting simple factual contradictions in a short story. A self-designed task of comparable scope may be proposed — see below.

Track B — Arithmetic Reasoning Specialist

Start from the course-provided general-language checkpoint and specialise it for arithmetic and arithmetic word-problem answering. The fixed task family covers ordinary arithmetic operations, direct numerical questions, and short application problems requiring both textual interpretation and calculation. The Project 3 repository's `test_pilot` file publishes the exact prompt/answer schema, supported operations and value ranges, required output format, representative wording, and difficulty range. Supervised instruction fine-tuning is the supported baseline; a short rationale or chain-of-thought-style response may be trained, but final-answer accuracy, format validity, and generalisation remain primary evidence.

Alternative Track A proposal and approval

The listed Track A capabilities may be used without separate approval. A different narrative specialisation of comparable scope needs approval before substantial training: submit the `proposal.md` template's frozen GitLab link through the Project 3 proposal panel by **Sunday 2 May 2027, 23:59 AET** (end of Week 9). It states the target behaviour, starting checkpoint, data and provenance, held-out evaluation, unchanged-model baseline and controlled comparison, one regression check, compute plan, and task-specific risks. The convenor responds within two teaching days with **approved**, **revise**, or **out of scope**. Before approval, only preparation and a pipeline dry-run of at most 0.5 T4-equivalent GPU-hours are permitted, and this still counts toward the Project 3 budget.

Marking breakdown

Component	Marks
Engineering report	20
Submitted model	10
Total	30

Engineering report — 20 marks

1. Task definition and success criteria — 3 marks. One clearly bounded target behaviour, a justified scope, valid success measures, and a test plan including unseen cases.

2. Starting checkpoint and data design — 3 marks. A justified permitted checkpoint, clear training/development/test separation, appropriate examples and counterexamples, and documented provenance and formatting.

3. Implementation and reproducibility — 3 marks. A correct fine-tuning pipeline, verified target masking, declared hyperparameters and seeds, reproducible commands, and a submitted package consistent with the report.

4. Controlled comparison and ablation — 4 marks. Comparison with the unchanged starting model and at least one controlled intervention addressing a meaningful uncertainty (data balance, rationale format, loss masking, curriculum, or regularisation).

5. Evaluation, generalisation, and regression analysis — 5 marks. Task-appropriate held-out metrics, tests beyond memorised templates, representative qualitative cases, an error taxonomy, and measurement of at least one retained capability.

6. Communication and course-level reflection — 2 marks. Concise, properly attributed reporting and a specific reflection on what the result demonstrates about task adaptation under limited model, data, compute, and time.

Submitted model — 10 marks

Submission validity is a technical gate. A valid model receives up to **6⅔ marks** from the shared tutor-evaluation PPL rule and up to **3⅓ marks** from track-specific performance review.

7. Tutor-evaluation perplexity — 6⅔ marks. The shared formula applied to answer-or continuation-only reference-token-normalised PPL across all ten tutor prompts: $P \leq 25$ earns all 6⅔ marks; $25 < P < 50$ receives the published exponential decay; $P \geq 50$ earns zero for this criterion.

8. Track-specific capability, generalisation, and reliability — 3⅓ marks.

Track A weights: objective satisfaction of the declared condition **40%**; narrative quality **25%**; held-out generalisation and prompt robustness **20%**; retention of the monitored starting capability **15%**. Track B weights: exact final-answer accuracy **55%**; required output-format validity **15%**; accuracy on unseen values and templates within the published scope **20%**; retained language and instruction reliability **10%**. Track B's first three dimensions are computed by the evaluation harness; blind raters assess only the qualitative dimension.

Individual work

This project is submitted individually. AI assistance is permitted when declared in the required AI Assistance Statement; responsibility for the submitted work remains with you. See [Policies](#) for the full individual-work, collaboration, and AI-assistance rules.

The spec

The brief poses the problem and leaves room for a response. The spec is the fixed part: what a marker checks. Some lines can be checked mechanically; the rest call for judgement.

- one specialisation track is chosen and one primary target behaviour is declared with measurable success criteria
- the starting checkpoint is declared, justified, and (when a personal earlier submission) verified against it
- the unchanged starting model is used as a baseline and at least one controlled comparison or ablation is run
- training and evaluation data are strictly held out from each other
- at least one relevant pre-existing capability is checked for regression
- an alternative Track A proposal, if used, was approved via proposal.md before substantial training
- training and evaluation stay within 12 T4-equivalent GPU-hours and 1.5×10^{17} training FLOPs
- Track B always starts from the course general-language checkpoint

Resources

START HERE

Full Project Brief

Printable copy of the common task, both tracks, constraints, evaluation rules, and rubric.

Will be available at 26 April 2027 at 9:00 am AET

Project 3 Starter Repository

Assigned fine-tuning code, proposal template, task schemas, test_pilot, budget ledger, and submission manifest.

Will be available at 26 April 2027 at 9:00 am AET

DATA AND MODEL

Starting Model and Task Data

Track-specific fallback checkpoint and data cards; Track B includes

Fine-tuning Starter Pack

Supported SFT Colab, response-only masking checks, example task records, and baseline commands.

the general-language checkpoint and arithmetic data.

Will be available at 26 April 2027 at 9:00 am AET

Will be available at 26 April 2027 at 9:00 am AET

EVALUATE

Track Evaluation Pack

Five development examples, ten tutor inputs, separation validator, task verifier, and regression utilities.

Will be available at 26 April 2027 at 9:00 am AET

SUBMIT

CVPR Report Template

Shared two-column LaTeX template with the required engineering-report and AI Assistance Statement sections.

Will be available at 26 April 2027 at 9:00 am AET

Submission

Will be available at 26 April 2027 at 12:00 am AET.

Due: 23 May 2027 at 11:59 pm AET

RELATED

[Project 2 – Give the Model a Voice](#)